

Physical Pharmacy

Code: (CPT 203/PT 203)

Lecture 1

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الرؤية لكلية الصيدلة الجامعة الحديثة:

"تتطلع كلية الصيدلة-الجامعة الحديثة للتكنولوجيا والمعلومات إلى التميز والابتكار في مجالات التعليم الصيدلي والبحث العلمي وخدمة المجتمع وتنمية البيئة محليا وإقليميا تحت مظلة أهداف ورؤية مصر ٢٠٣٠ للتنمية المستدامة"

الرسالة لكلية الصيدلة الجامعة الحديثة :

"تلتزم كلية الصيدلة بالجامعة الحديثة للتكنولوجيا والمعلومات بتقديم برامج أكاديمية متطورة في التعليم الصيدلي لأعداد خريج متميز قادر علي تطوير منظومة الرعاية الصحية والارتقاء بالخدمات الصيدلية والتصنيع الدوائي وتحقيق مشاركة فعالة في مجال البحث العلمي وخدمة المجتمع لتحقيق أهداف التنمية المستدامة في إطار من القيم والأخلاقيات."

Overall aims of the course:

- This course provides students with knowledge of physicochemical principles essential for the design and formulation of pharmaceutical products. Students are introduced to the fundamental concepts of states of matter, Phase equilibrium, colligative properties, is tonicity solubility, dissolution, partition coefficient, surface and interfacial phenomena, surface active agents, adsorption and its application in pharmacy and rheological behavior of dosage forms.

Student assessment methods:

Methods of Assessment	Marks
Midterm	25
Final written Exams	50
Practical	40
Quizzes	10
Assignments	10
Oral	15
Total	150

Lecture 1

State of matter and solution

State of matter

Solid

Liquid

Gas

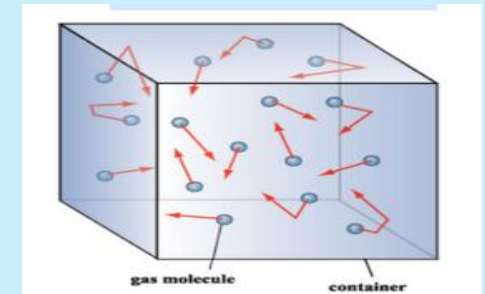
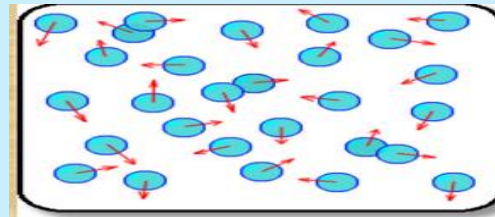
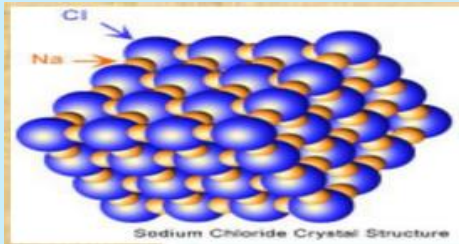
1. Type of bonds

- Van der Waal forces
- Hydrogen bonding
- Covalent and ionic forces

Intermolecular Forces

- Van der Waal forces
- Hydrogen bonding

-Non-existent



2. Shape

Fixed

Not Fixed

Not Fixed

3. Volume

Fixed

Fixed

Not defined

State of matter

Solid

Particles are immobile.
They can't move freely
but can only vibrate.

Liquid

Molecules are
mobile in the three
axes

Gas

Molecules move in
straight paths in all
directions at high
velocities.

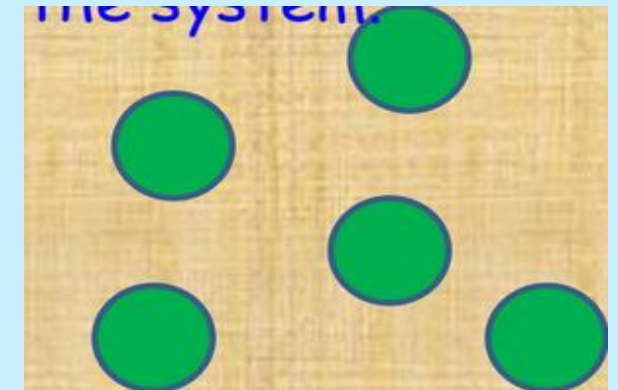
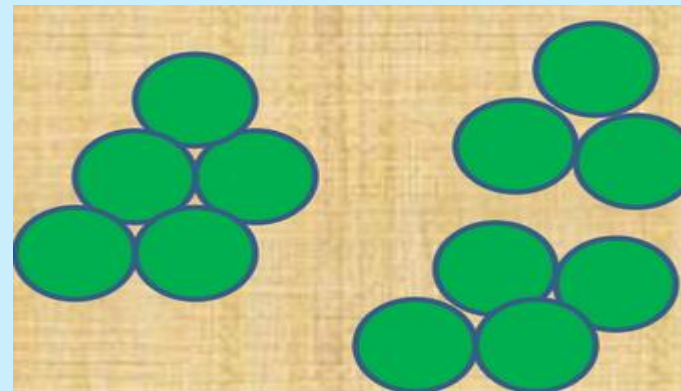
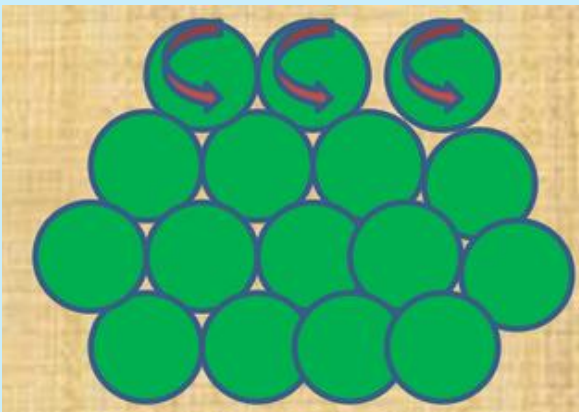
4. Mobility

5. Interspaces

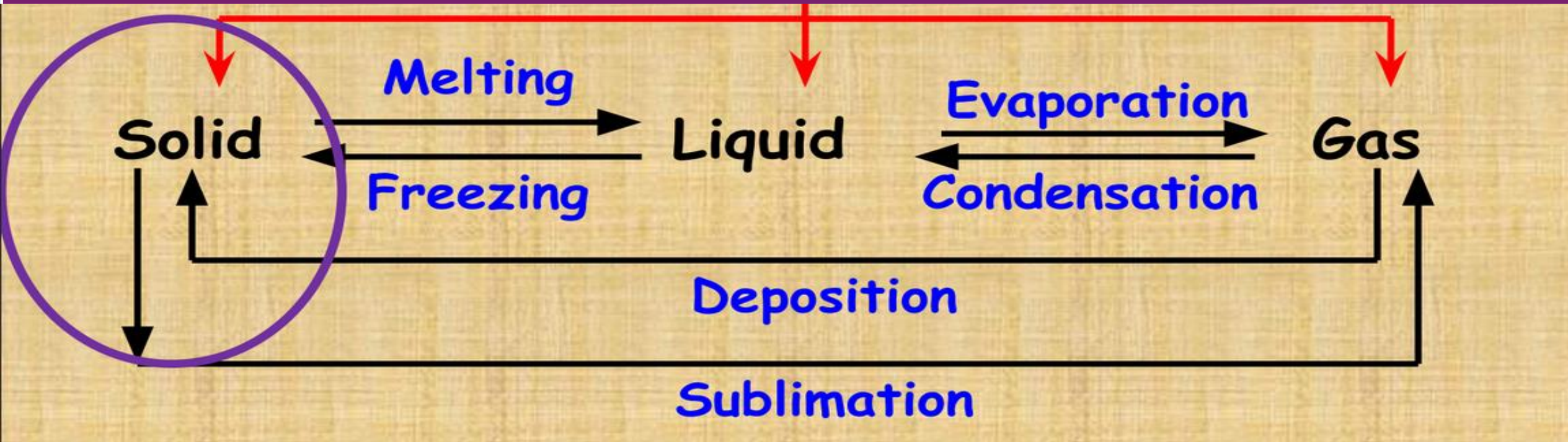
Ver small

Moderate

High



Process of changing the State of matter



1. **Melting** (الانصهار): Transferring of solid state to liquid state by increasing the temperature.
2. **Freezing** (التجمّد): Transferring of liquid state to solid state by decreasing the temperature.
3. **Evaporation** (التبخر): Transferring of liquid state to gas state by increasing the temperature.

Process of changing the State of matter

4. Condensation (التكثيف): Transferring of gas state to liquid state by decreasing the temperature Freezing .

5. Sublimation: Conversion from the solid to the gaseous state without melting
e.g. Iodine

عملية التسامي: هي عملية التسخين السريع جدا لدرجة تحول المادة الصلبة لمادة غازية دون انصهار أو بدون المرور بالحالة السائلة (تحدث لبعض المواد فقط)

6. Deposition: Recondensation of gas into solid state.

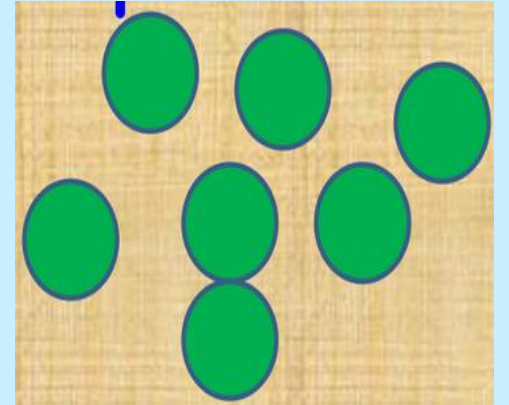
There is a fourth state of matter: **Liquid Crystal** ; كريستالات او زجاج مكسور
اعادة التكثف: تحدث للمواد الغازية ولكن تتكون لصورة صلبة دون المرور بالحالة السائلة، وتحدث بسبب التبريد الشديد لبعض المواد الغازية فتكون حالة خاصة جديدة من حالات المادة (البلورات)

Solid State

Crystalline Solid



Amorphous Solid



مش كل المواد الصلبة لها نفس الترتيب الداخلي

Solid State

Crystalline Solid

-Structural units are arranged in fixed geometrical patterns or lattice

الماس و الملح

-e.g. cubic → NaCl

-More stable

Amorphous Solid

-Molecules are arranged in a random manner (considered supercooled liquids).

الزجاج والبلاستيك

e.g. Glass



e.g. Novobiocin

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graph TD; A[e.g. Novobiocin] --> B[Crystalline Solid]; A --> C[Amorphous Solid]; B --> D["-More Stable<br/>-poorly absorbed has poor activity"]; C --> E["-Less stable (contains high energy)<br/>-More Soluble<br/>-well absorbed<br/>-Therapeutically active."];
```

Crystalline Solid

- More Stable**
- poorly absorbed has poor activity**

Amorphous Solid

- Less stable
(contains high energy)**
- More Soluble**
- well absorbed**
- Therapeutically active.**

Crystalline Solid

consists of units that are either:

Ions

NaCl

Atoms

**Diamond and
Graphite**

Molecules

Naphthalene

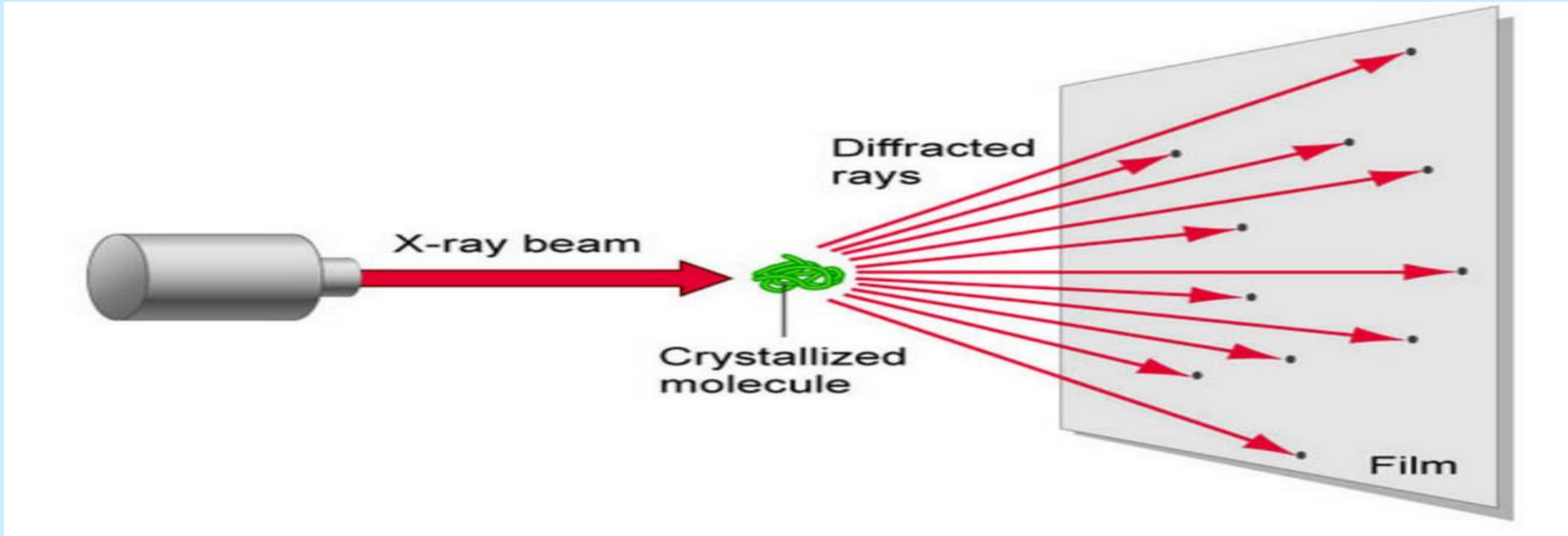
Crystalline Solid Characterization

A. X-Ray Diffraction

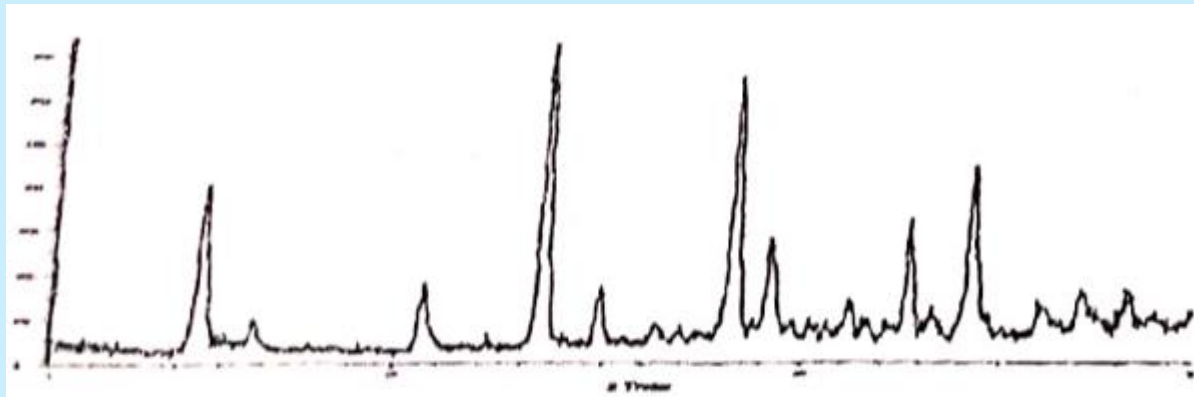
B. Melting point and heat of fusion

C. Polymorphism

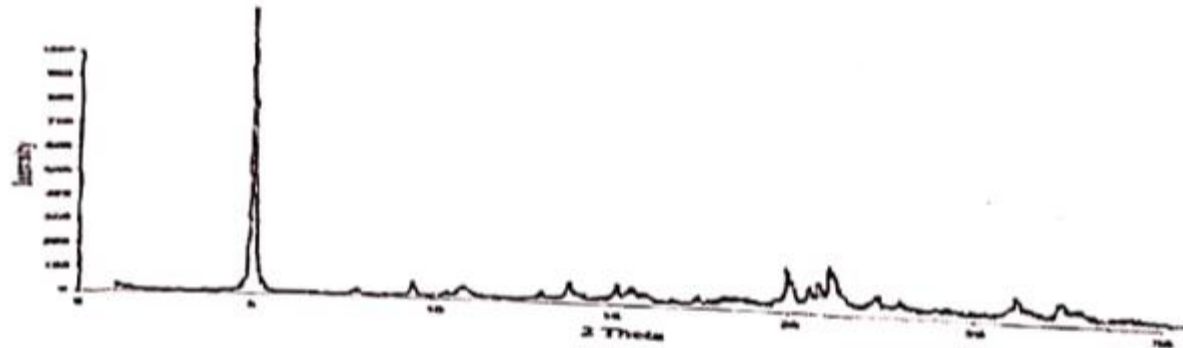
A. X-Ray Diffraction



Reflection of X-ray beam from atomic planes of crystal → structure of crystal is investigated.



Crystalline



Amorphous

Crystalline Solid Characterization

B. Melting point and heat of fusion

Latent Heat of Fusion :

➤ Heat absorbed when a gram of a solid melts

Or

➤ Heat liberated when a gram of liquid freezes

- A crystal that is bound by weak forces  Low heat of fusion
Low melting point
- A crystal that is bound by strong forces  High heat of fusion
High melting point

Crystalline Solid Characterization

C. Polymorphism

Polymorphism occurs when the molecules arrange themselves in two or more different ways in the crystal.

Polymorphs: Substances existing in more than one crystalline form.

They are chemically identical but differ in melting points, X-ray diffraction patterns, and solubility.

e.g. Carbon (Diamond & Graphite)

Ice: Ice has 15 known crystal structures which exist at various temperatures and pressures.

Cocoa butter (We should use only stable β form)

Liquid crystalline State

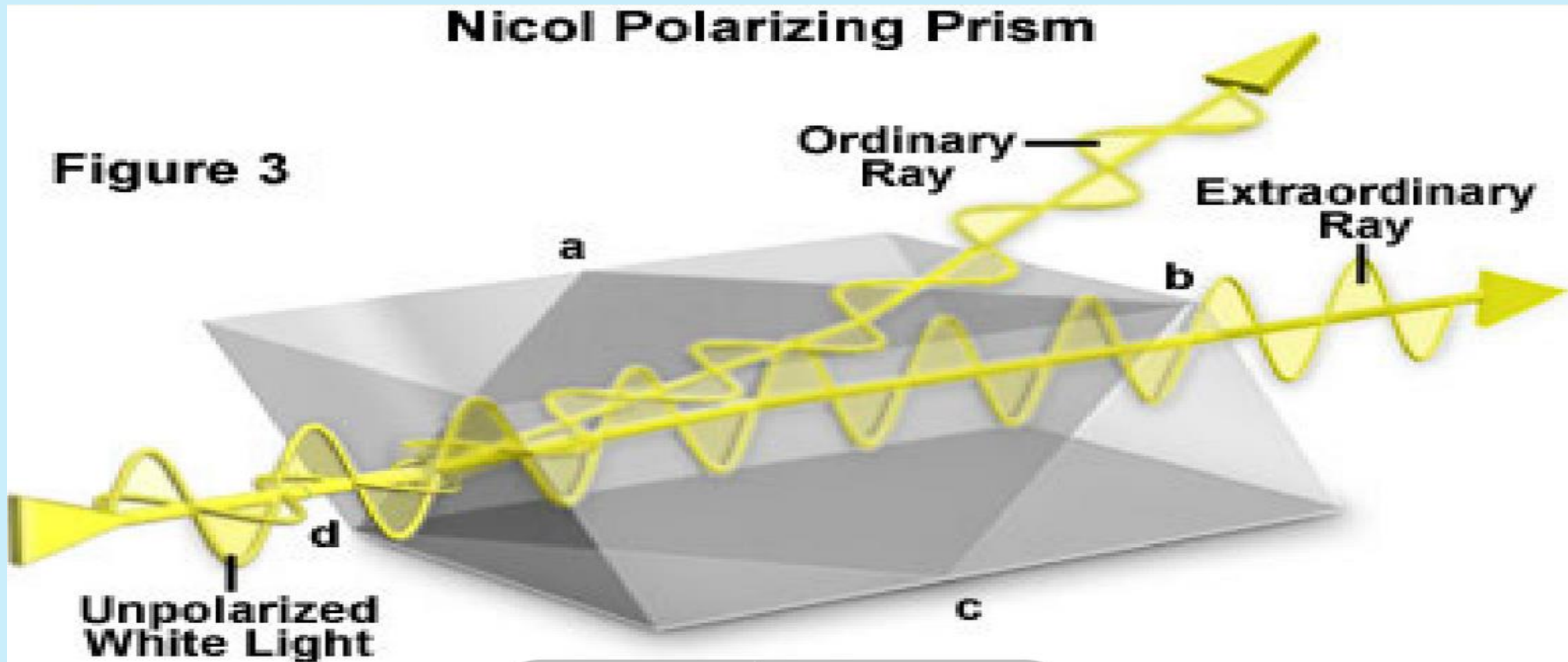
Fourth state of matter

Intermediate between the liquid and solid states called **mesophase**.

Meso = Intermediate

- **Properties of liquid crystals:**
- ✓ **Some of liquid properties** : Mobile and have the flow properties of liquids.
- ✓ **Some of solid properties**: Birefringent (a property of crystals → light passing through material is divided into two components with different refractive indices).

Birefringent



Liquid crystalline State

Liquid crystal may result from:

Heating of solid



Thermotropic
liquid crystals

- The first thermotropic liquid crystal was made by Friedrich by heating solid cholesteryl benzoate at 145°C → turbid liquid (liquid crystal)
- At 179°C it became clear (conventional liquid state)

Action of solvents
on solids

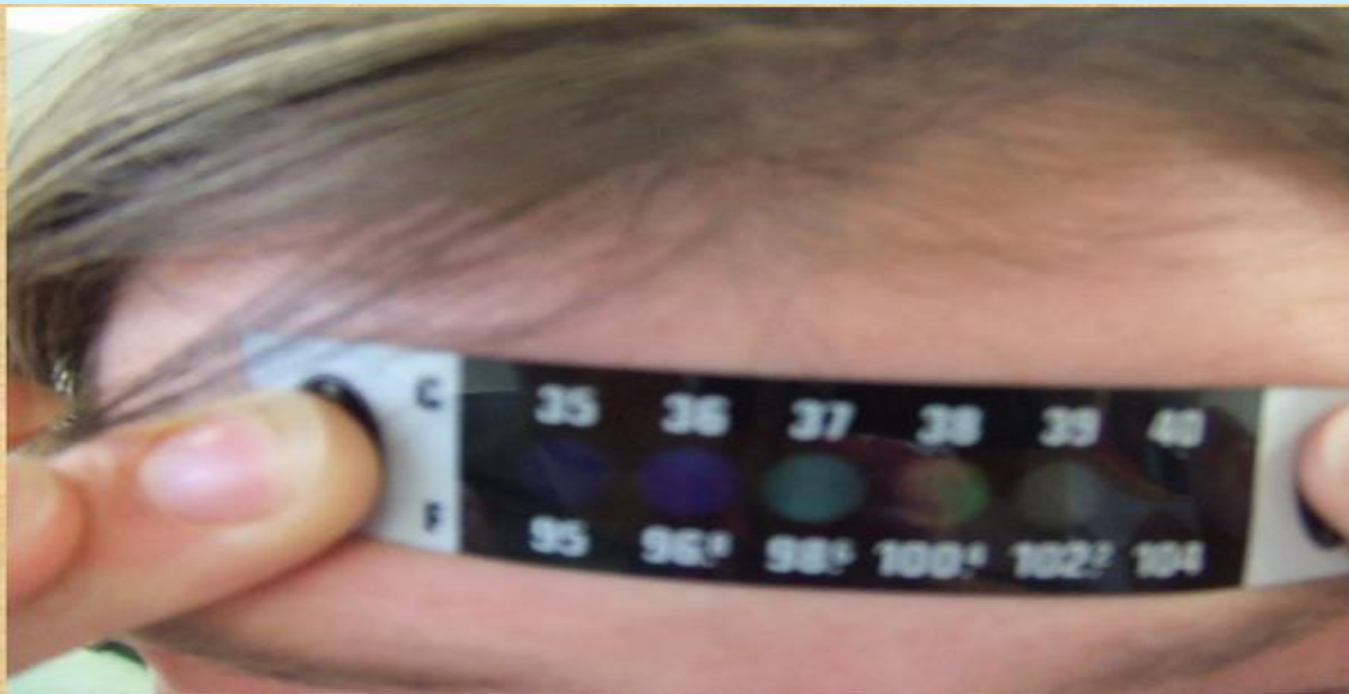


Lyotropic liquid
crystals

Liquid crystalline State

Significance and Uses of Liquid crystals:

Some liquid crystals show color changes with temperature used to detect areas of elevated temperature under skin that may be due to a disease process.



Liquid crystalline State

Significance and Uses of Liquid crystals:

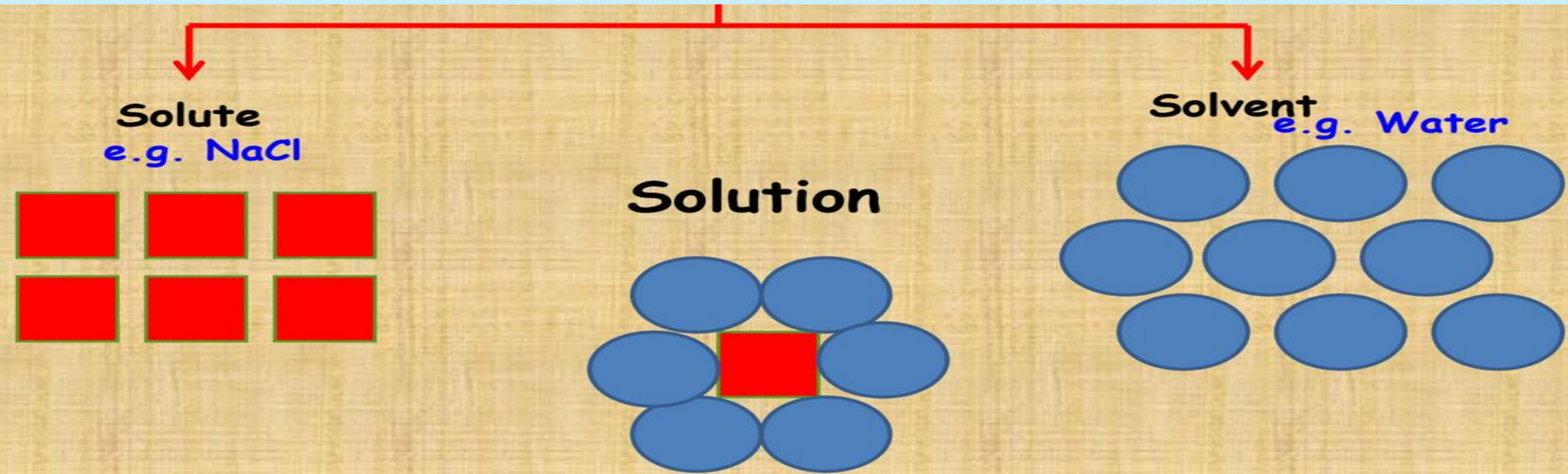
Some liquid crystals are sensitive to electric fields used in developing display systems (LCD).



Solutions and their properties

Solution

- ✓ Mixture of two or more components that form single phase.
- ✓ The component which is to be dissolved is called solute.
- ✓ The component in which the solute is dispersed to get a homogenous mixture is called the solvent.



How Does a Solution Form?

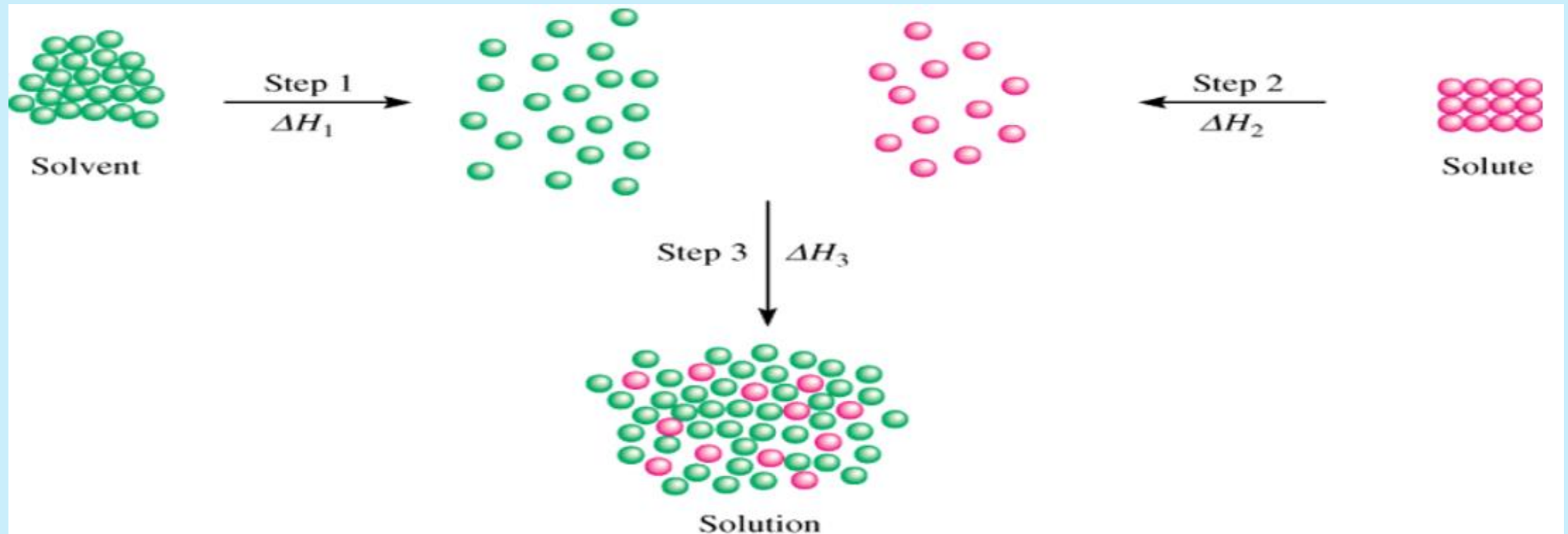
There are three intermolecular forces between solution components:

- **Solvent-solvent** (Cohesion: the attraction of like molecules)
- **Solute –solute** : (Cohesion)
- **Solvent-solute** : (Adhesion : the attraction of unlike molecules)

Simply, three processes affect the solution formation:

- Separation of solute particles
- Separation of solvent particles
- New interactions between solute and solvent.

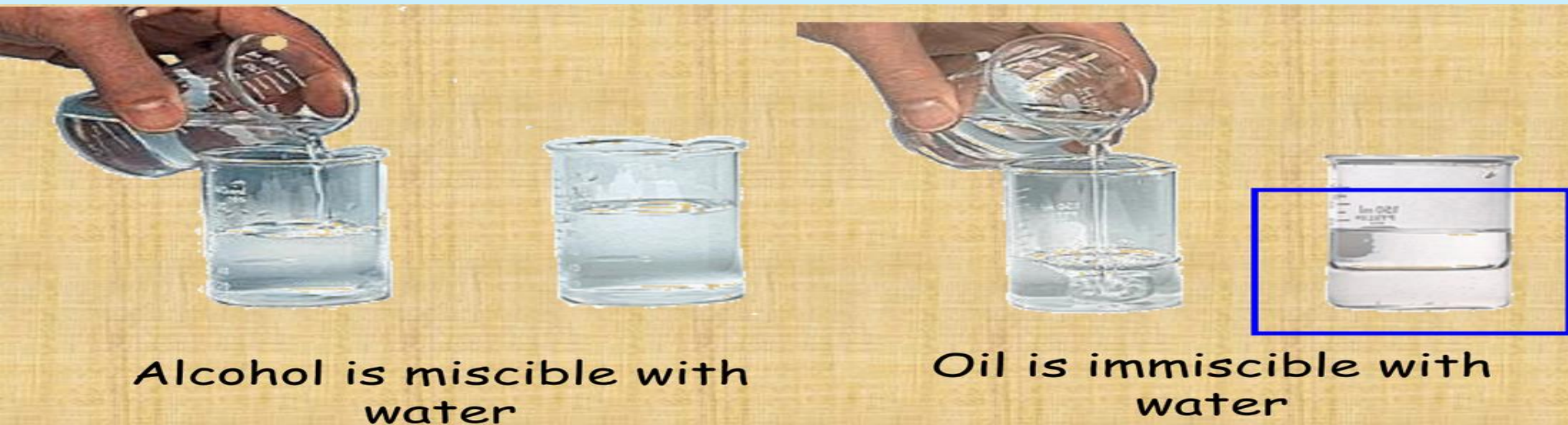
- ❖ Solvent-solvent interaction (endothermic: ie: energy required)
- ❖ Solute-solute interaction (endothermic)
- ❖ Solvent-solute interaction (exothermic: ie: energy released)



Solutions and their properties

If the substance dissolves it will be called **soluble** and if not it will be **insoluble**.

- **Miscible** means soluble without limits and **immiscible** means insoluble.
- **For example, alcohol is miscible with water.**
while oil is immiscible with water.



Types of Solutions

1)Unsaturated Solution 2)Saturated Solution 3)Supersaturated Solution

Unsaturated Solution :The dissolved solute is present in a **concentration below** that necessary for complete saturation at definite temperature.

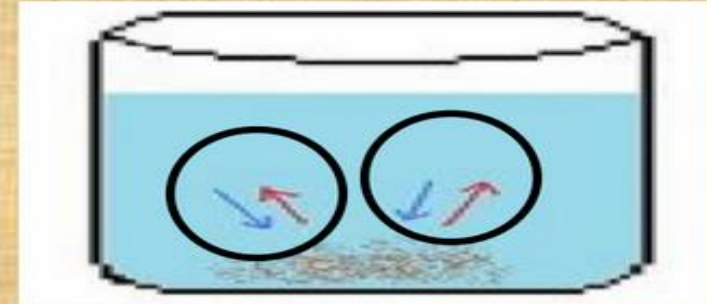
Saturated Solution :The solution is in **dynamic equilibrium** with excess undissolved solid at a **definite temperature**.

Supersaturated Solution :Contains more dissolved solute than it would normally contain at a definite temperature.

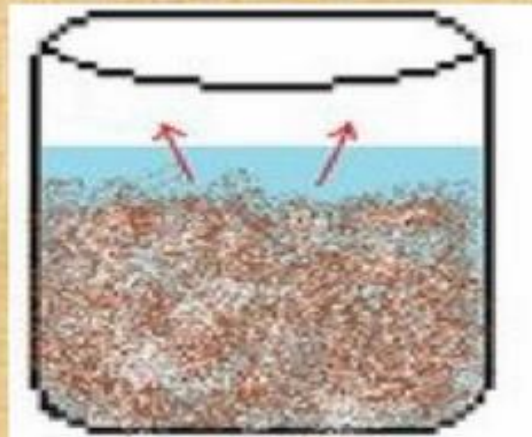
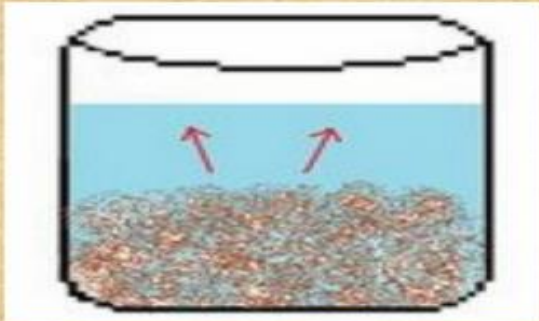
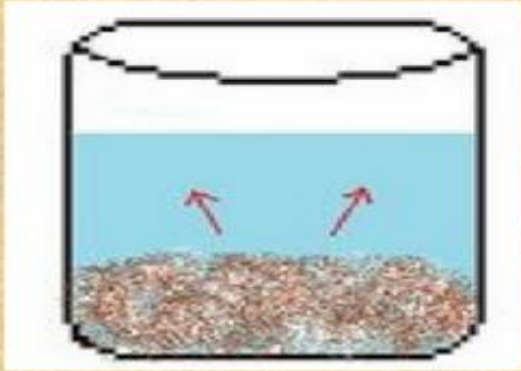
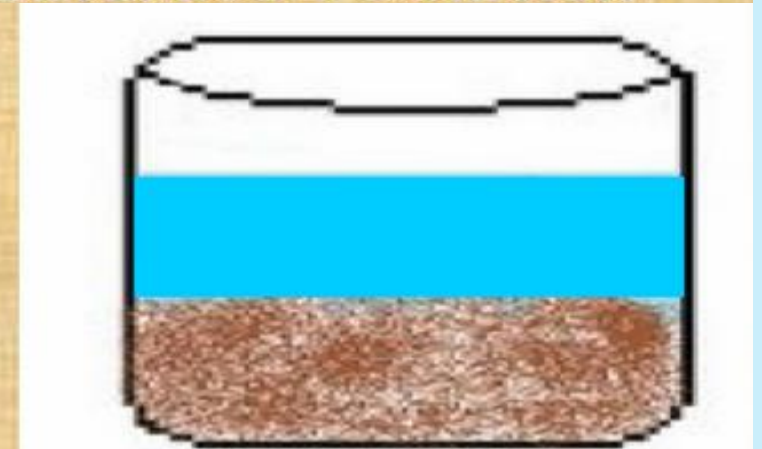
Unsaturated Solution



Saturated Solution



Supersaturated Solution



Methods For Determination of Solubility



**Analytical
Method**

**Synthetic
Method**

**Modified
Synthetic
Method**

**Phase
Solubility
Analysis**

References :

- Florence, Alexander T; Attwood, David 2015, Physicochemical principles of pharmacy Australian prescriber.
- Blaine Templar Smith 2016 Remington: The Science and Practice of Pharmacy Pharmaceutical Press.
- Ram I. Mahato, Ajit S. Narang, 2017 Pharmaceutical Dosage Forms and Drug Delivery 3rd ed, Pharmaceutical Press

Thank you

The image features the words "Thank you" in a 3D isometric font. The letters are light yellow with blue outlines and are arranged in two rows: "Thank" on top and "you" below it. Each letter casts a soft red shadow onto the pink background. Scattered around the text are several colorful, five-pointed stars in shades of cyan, yellow, and purple, each also casting a red shadow. The overall style is bright, cheerful, and modern.